

## Lab worksheet 4: Defining Classes

### Instructions

1. Create a folder and name it using your student number in the format "**CT2022XXX**". (Eg: **CT2022001**)
2. Create a Java project in IntelliJ inside your folder and name it using the Lab worksheet number in the format "**LW\_XX**". (Eg: **LW1**)
3. Create separate Packages for each question and name them with their question number in the format "**QXX**". (Eg: **Q1**)
4. Create a repository in your GitHub and name it using your student number and upload your project files.

### Questions

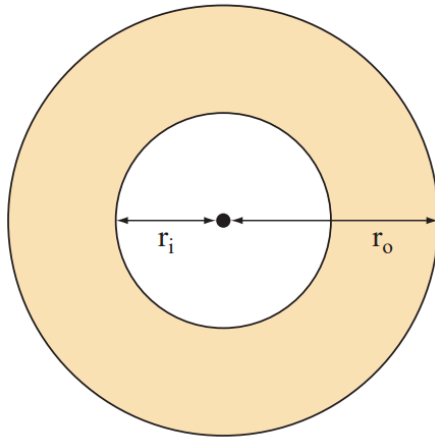
1. Define a new class named **Temperature**. Save the temperature internally in Celsius using a private double variable **celsius**. The class has two Constructors one is a No-Arg Constructor, and another one is Parameterized Constructor which accepts a double type argument celsius. The class also has two getter/accessor methods, **toFahrenheit** and **toCelsius** that return the temperature in the specified unit, and two setter/mutator methods **setFahrenheit** and **setCelsius** that assign the temperature in the specified unit. Using this class, write a Java program that accepts temperature in Celsius as an input using the Scanner class and outputs the temperature in equivalent Fahrenheit.

**celsius = (fahrenheit - 32) \* 5 / 9**

**fahrenheit = celsius \* 9 / 5 + 32**

2. Using the **setFahrenheit** method of **Temperature** class from the above Exercise, write a Java program that accepts temperature in Fahrenheit as an input using the **Scanner** class and outputs the temperature in equivalent Celsius.
3. Write a Java program that computes the area of a circular region (the shaded area in the diagram), given the radii of the inner and the outer circles,  $r_i$  and  $r_o$ , respectively. We compute the area of the circular region by subtracting the area of the inner circle from the area of the outer circle. Define a **Circle** class that has methods **computeArea** and **computeCircumference** to compute the area and circumference. You set the circle's radius with the **setRadius** method or via a constructor.

## Lab worksheet 4: Defining Classes



4. Modify the **Bicycle** class provided below, to assign an owner object instead of just the owner's name and telephone number. The new **Owner** class will have two data members **ownerName** and **phoneNo**.

File: **Bicycle.java**

```
class Bicycle {
    // Data Member
    private String ownerName;
    private String phoneNo;

    //Constructor: Initializes the data member
    public Bicycle() {
        ownerName = "Unknown";
    }

    public Bicycle(String name,String num) {
        ownerName = name;
        phoneNo = num;
    }

    //Returns the name of this bicycle's owner
    public String getOwnerName() {
        return ownerName;
    }

    //Assigns the name of this bicycle's owner
    public void setOwnerName(String name) {
        ownerName = name;
    }
}
```

## Lab worksheet 4: Defining Classes

```
    }

    public String getPhoneNo() {
        return phoneNo;
    }

    //Assigns the name of this bicycle's owner
    public void setPhoneNo(String num) {
        phoneNo = num;
    }
}
```

5. Please write a Java program for a university course registration system. The program should consist of the following classes: **Course**, **Lecturer**, **Student**, and **Main**.

In the **Course** class, include two String variables: **courseName** and **courseCode**. Additionally, include a **lecturer** object along with their corresponding getter and setter methods.

The **Lecturer** class should have two String variables named **lecturerName**, **courseTeaching** along with their getter and setter methods.

The **Student** class should include three String variables: **studentName**, **degreeName**, and **courseFollowing**. It should also have corresponding getter and setter methods.

Lastly, implement a **Main** class to serve as the entry point of the program. Create **course**, **lecturer**, and **student** objects, and assign them all the necessary properties.